

AI Agent Compliance Assessment Report

EU AI Act Article 12 + SOC 2 Trust Service Criteria

Assessment Period: January 01, 2026 - March 30, 2026

Report Generated: April 01, 2026

Data Source: Agent Observability Platform

Traces Analyzed: 50

Total Events: 303

Agent: research_writer_agent

Executive Summary

EU AI Act Article 12: 83.3%

SOC 2 Trust Service Criteria: 83.3%

This assessment analyzed 50 agent traces containing 303 events over a 88-day period.

Of 21 total compliance requirements assessed, 15 are fully met, 5 are partially met, and 1 have critical gaps.

There are 1 critical gaps requiring immediate remediation and 5 warnings to address.

EU AI Act Article 12 Assessment

Requirement	Article	Status	Evidence / Gap									
Automatic Recording of Events	Art. 12(1)	MET	Evidence: 50 items									
Risk Event Recording	Art. 12(2)(a)	MET	Evidence: 5 items									
Post-Market Monitoring Support	Art. 12(2)(b)	MET	Evidence: 7 items									
Operational Monitoring	Art. 12(2)(c)	MET	Evidence: 50 items									
Session Timestamps	Art. 12(3)(a)	MET	Evidence: 50 items									
Input Data Recording	Art. 12(3)(c)	MET	Evidence: 248 items									
Human Oversight Logging	Art. 14	PARTIAL	Only 15 of 50 traces (30%) contain human oversight events. Article 14 requires logging									
Log Retention (6 months)	Art. 26	GAP	Trace data spans 88 days. Article 26 requires minimum 6 months (180 days) of log retention									
Incident Reporting Support	Art. 72/79	MET	Evidence: 5 items									
Component Identification	EN 18229-1	MET	Evidence: 303 items									
Timestamp Completeness	EN 18229-1	MET	Evidence: 303 items									
Tamper Evidence / Log Integrity	Implied / EN 18229-1	PARTIAL	Only 30 of 50 traces (60%) have hash-chain integrity verification.									

SOC 2 Trust Service Criteria Assessment

Requirement	CC Code	Status	Evidence / Gap									
Anomaly Detection & Monitoring	CC7.1	MET									Evidence: 5 items	
Failure Monitoring	CC7.2	MET									Evidence: 5 items	
Event Triage & Severity Classification	CC7.3	MET									Evidence: 5 items	
Root Cause Analysis Support	CC7.5	MET									Evidence: 5 items	
Human Oversight Policies Deployed	CC5.3	PARTIAL									Human oversight policies evidenced in only 30% of traces	
Change Management (Version Logging)	CC8.1	PARTIAL									Evidence: 150 items	
Third-Party Communication Logging	CC2.3	MET									Evidence: 33 items	
Input Completeness	PI1.3	MET									Evidence: 248 items	
Output Completeness	PI1.4	PARTIAL									Evidence: 198 items	

Gap Analysis & Remediation Plan

[CRITICAL] Log Retention (6 months) (Art. 26) - EU AI Act

Current: Trace data spans 88 days. Article 26 requires minimum 6 months (180 days) of log retention.

Remediation:

- Configure your observability tool to retain data for at least 6 months
- Langfuse Cloud: check your plan's retention limits
- Self-hosted: configure database retention policies
- For financial services: align with your existing record-keeping obligations

[WARNING] Human Oversight Logging (Art. 14) - EU AI Act

Current: Only 15 of 50 traces (30%) contain human oversight events. Article 14 requires logging of human intervention points for high-risk AI systems.

Remediation:

- Add human-in-the-loop checkpoints to your LangGraph agent using `interrupt_before/interrupt_after`
- In Langfuse: use `trace.event(name='human_oversight', metadata={reviewer_id: ..., decision: ...})`
- In Laminar: add `@observe` decorator on approval/review functions
- Ensure every high-risk decision path includes a human review gate

[WARNING] Tamper Evidence / Log Integrity (Implied / EN 18229-1) - EU AI Act

Current: Only 30 of 50 traces (60%) have hash-chain integrity verification.

Remediation:

- Enable hash-chain logging if your observability tool supports it
- Alternatively, export traces to an append-only store (e.g., S3 with Object Lock)
- Consider adding cryptographic signatures to trace batches

[WARNING] Human Oversight Policies Deployed (CC5.3) - SOC 2

Current: Human oversight policies evidenced in only 30% of traces

Remediation:

- Same as Article 14: add human oversight checkpoints and log them
- Document your human oversight policy and reference it in trace metadata

[WARNING] Change Management (Version Logging) (CC8.1) - SOC 2

Current: Status: partial (150 evidence items)

Remediation:

- Review and address this criterion

[WARNING] Output Completeness (PI1.4) - SOC 2

Current: Status: partial (198 evidence items)

Remediation:

- Review and address this criterion

Methodology

Data Source: Agent observability trace data

Assessment Period: 2026-01-01 to 2026-03-30 (88 days)

Traces Analyzed: 50

Events Analyzed: 303

EU AI Act Requirements Assessed: 12 (Article 12 + connected articles)

SOC 2 Criteria Assessed: 9 (selected CC + additional criteria)

Assessment Rules: AgentAudit v0.1.0 rule set

Confidence Level: Medium (based on 50 traces)

This report was generated automatically by AgentAudit. The compliance assessment is based on analysis of agent trace data and should be reviewed by qualified legal and compliance professionals. This report does not constitute legal advice.